

Chapter 16.2: Vector Line Integrals

V1: Line Integrals. I can set up and evaluate scalar and vector field line integrals in two and three dimensions.

A2: Integral Applications. I can use multiple integrals to solve physical problems, such as finding area, average value, volume, or the mass or center of mass of a lamina or solid. I can interpret mass, center of mass, work, flow, circulation, flux, and surface area in terms of line and/or surface integrals, as appropriate.

Mechanics

1. Evaluate $\int_C (2x - y)\mathbf{i} \cdot d\mathbf{r}$ where C is parameterized by $\mathbf{r}(t) = (t^2)\mathbf{i} + (3t - 2)\mathbf{j}$, $t \in [0, 1]$.
2. Consider the closed curve C consisting of a semicircle and a straight line segment as follow:

$$\mathbf{r}_1(t) = (2 \cos t)\mathbf{i} + (2 \sin t)\mathbf{j}, \quad t \in [0, \pi], \quad \mathbf{r}_2(t) = t\mathbf{i}, \quad t \in [-2, 2]$$

Let the vector field $\mathbf{F}$ be given by

$$\mathbf{F}(x, y) = -y^2\mathbf{i} + x^2\mathbf{j}.$$

Find the circulation of $\mathbf{F}$ around C and the flux of $\mathbf{F}$ across C .

3. Find the work done by the force $\mathbf{F} = xy\mathbf{i} + (y - x)\mathbf{j}$ over the straight line from $(1, 1)$ to $(2, 3)$.

Applications

4. *Ampere's Law* states that for a magnetic field $\mathbf{B}$, and closed curve C , we have that

$$\int_C \mathbf{B} \cdot d\mathbf{r} = \mu_0 I_{enc}$$

where μ_0 is a physical constant, and I_{enc} is the total current in the region enclosed by C . Compute I_{enc} in the unit square $[0, 1] \times [0, 1]$ induced by the magnetic field $\mathbf{B}(x, y) = (x^2y, 5 - xy^2)$. Also compute the magnetic flux across the same square.

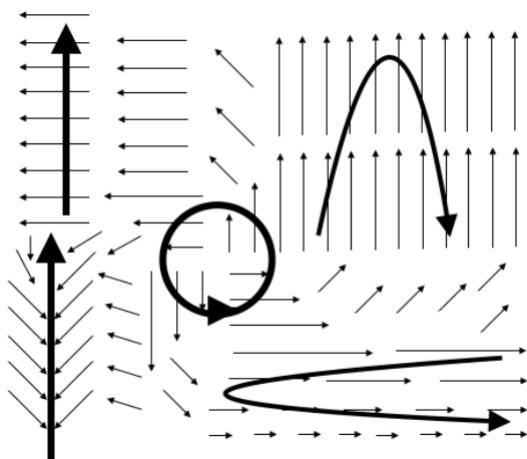
5. Consider a radial force field $\mathbf{F}(x, y) = (Kx, Ky)$ for $K > 0$, and a particle moving counterclockwise on the unit circle. How much work does the force field exert on the particle over one loop around the circle? Give an example of a real life force whose field look like $\mathbf{F}$ (up to choosing a constant K).

Extensions

6. Consider the vector field $\mathbf{F}$ (thin arrows) and let $\mathbf{T}$ denote the unit tangent vector to the oriented curves shown below (thick arrows). Determine whether

$$\int_C \mathbf{F} \cdot \mathbf{T} \, ds$$

is positive, negative, or zero for each oriented curve C . In other words, determine whether the work done by the vector field on each curve is positive, negative, or zero.



7. The *Work-Energy Theorem* tells us that the total work done by a *conservative* force field $\mathbf{F}$ equals the difference in potential energy:

$$\int_C \mathbf{F} \cdot \mathbf{T} \, ds = PE_f - PE_i$$

On the other hand, the principle of conservation of energy tells us that when we traverse any closed path, no (potential) energy can be gained or lost; it must always be preserved. Using only this fact, verify that the following vector fields are not conservative force fields by integrating over an appropriate loop:

(a) $\mathbf{F}(x, y) = (x^2y, 5 - xy^2)$

(b) $\mathbf{F}(x, y) = (xy, x^2)$

(c) $\mathbf{F}(x, y) = \left(\frac{-y}{x^2+y^2}, \frac{x}{x^2+y^2} \right)$ [Hint: Those fractions would look a lot nicer if their denominators were equal to 1.]